

and applications, understanding where automation fits—from complex processes to simple pick-and-place tasks.

We also emphasise end-of-arm tools, such as glue dispensers, which help students understand precision, nozzle selection, and material control. This hands-on exposure helps bridge the gap between theory and practice and gives students a clearer understanding of real industrial automation challenges.

How is the programme's impact on student employability and industrial productivity measured?

We measure success by how effectively students are prepared to enter the automation industry.

We collect students' CVs and share them with system integrators and other companies involved in factory automation, creating a reference pool for potential employers. We also provide participation certificates that students can use when applying for roles.

This approach positions them as job-ready candidates with foundational knowledge of factory automation. By supplying industry-ready talent, we help reduce the training burden on companies while accelerating the adoption of factory automation in India.

How are advanced robotic labs funded, maintained, and kept industry-relevant?

Initially, we provided robots to partner universities at no cost, while the institutions supplied space and basic infrastructure.

As the programme expanded, investment became a collaborative effort, covering robots as well as safety systems, vision cameras, force sensors, and feeder units. Maintenance and upgrades are jointly managed to ensure the labs remain current and industry-relevant.

Typical equipment includes SCARA and six-axis robots, vision systems, force sensors, and automation accessories integrated to support hands-on learning and real industrial applications.

How many students use the robotics lab at a time, and how is it structured for industry-academia collaboration?

At our labs in Bengaluru and Chennai, we can accommodate around

30 students at a time.

This structure enables focused, hands-on training while supporting industry-academia collaboration. Students work closely with instructors and, where applicable, industry experts, gaining practical experience that complements their academic learning.

How do students access robotic labs, and how is meaningful engagement ensured?

Students can access our labs at no cost and are encouraged to explore projects.

Beyond lab access, we provide on-site training, integrate real research and development projects, and organise periodic seminars to share industry trends. Continuous engagement helps ensure students stay updated on emerging technologies and gain meaningful, industry-relevant experience.

With 100+ students benefiting in FY2025, how will the initiative be scaled?

We plan to scale the initiative by partnering with more institutions and deepening engagement with selected colleges. We are also exploring collaboration with industry associations, manufacturing clusters, and government bodies to strengthen the robotics ecosystem in India.

Currently, training is delivered by our sales engineers as part of their existing roles. However, as demand grows, we are considering establishing a dedicated training team focused solely on this initiative.

At the same time, we are expanding robotics labs across more universities nationwide. This will enable students to gain practical skills and help bridge the gap between robotics graduates and industry-ready engineers.

What collective initiatives can robotics companies take to build a globally competitive automation workforce in India?

Knowledge sharing is the most critical initiative.

Companies need to demonstrate that automation is neither overwhelming nor excessively costly. Many MSME customers believe factory automation requires large investments

or a complete redesign of production lines.

However, operations can often be automated using simple, affordable machines with straightforward designs. Sharing expertise and practical guidance encourages industries to adopt automation more quickly and confidently.

Are you planning to expand to more colleges?

Epson currently operates lab setups in Bengaluru and Delhi, and is exploring engagement with more students from engineering and diploma colleges.

Students visit our labs for hands-on training, and we cover basic expenses such as lunch and small appreciation incentives to encourage participation. These costs are currently funded by Epson.

As the programme grows and more students participate, we will evaluate the investment model to sustain and scale engagement. In Delhi, the lab is located within our office.

How will robotics adoption in India evolve over the next five years, and what role can global leaders play in strengthening the Make in India ecosystem?

Over the next five years, we expect automation adoption in India to mature significantly. Many companies will gradually shift manual operations towards automated solutions, improving precision, speed, and efficiency.

At Epson, our focus is on compact robots designed for high precision and fast operation, supported by our RC Plus software. This software allows users to program robots, simulate processes in 3D, and control systems more easily, making automation accessible even to teams with limited programming knowledge.

Global technology leaders can support this ecosystem by enabling knowledge transfer, collaborating with Indian manufacturers, and contributing to local innovation and scalable adoption of advanced robotics. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/IntEfyApril1>

